

Three-dimensional data of wire-cut surface scans under the confocal microscope

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Abstract

[Agent: For a Data Descriptor, revise the abstract sentence reporting higher same-source correlations and decreasing variability so it describes the data and reuse without making a claim about a scientific finding.]

Wire-cut striation marks can provide evidence for forensic source identification, but publicly available datasets and reproducible quantitative workflows for analyzing these marks are limited. We present 120 three-dimensional topographic scans in `x3p` format from aluminum wires cut with five KWS-105 wire cutters. The study includes four edges, three locations along each tool, and two replicates for every tool-edge-location combination. We provide scan metadata, manually extracted profiles, processed signals, pairwise cross-correlation values, images, and analysis code. The accompanying workflow covers scan inspection, profile and signal extraction, cross-correlation-based alignment, pairwise similarity visualization, and variability assessment across replicates, stagings, and acquisitions. Technical validation shows higher cross-correlations for same-source comparisons than for different-source comparisons and decreasing variability as scanning conditions become more consistent. The dataset and workflow support the development, evaluation, and reproducible comparison of quantitative methods for wire-cut evidence.

Background & Summary

In forensic analysis, marks left at crime scenes play an important role, and forensic examiners are particularly interested in identifying their origin, a process known as the Source Identification Problem in Forensic Science¹. The forensic science community focuses on two distinct problems: the **specific source problem**, in which examiners aim to determine whether a mark was left by a particular tool, and the **common source problem**, in which examiners aim to determine whether two marks were left by the same tool. Currently, accepted practice for both problems relies on visual inspection of physical items under a comparison microscope. Examiners then report their conclusions according to the Theory of Identification², developed by the Association of Firearm and Toolmark Examiners (AFTE), using one of three categories: *identification*, in which the marks are believed to have been made by the same source; *elimination*, in which the marks are believed to have been made by different tools; and *inconclusive*, in which the similarities or dissimilarities between marks are insufficient to support either identification or elimination. This process has been criticized by the National Research Council (NRC)³ and the President's Council of Advisors on Science and Technology (PCAST)⁴ for its lack of objectivity and absence of quantifiable measurements, such as error rates.

In discussing error rates, Biedermann⁵ differentiates between internal and external perspectives. The external perspective allows general statements based on black-box studies that relate examiners' conclusions to ground truth; however, it does not consider the evidence in a particular case. Introducing an internal perspective requires a quantitative evaluation of the similarity of specific evidence and, in particular, objective algorithmic evaluations.



Figure 1. Microscopic wire-cut striations. Close-up of striations left by a blade on the cut end of a wire.

To train algorithms, we need data from scenarios with known ground truth. Because data collected at crime scenes are not suitable for this purpose, we used specific tools to cut wires under controlled conditions and collected the resulting marks for training. This approach enables comparisons between ground truth and the classification results produced by trained algorithms, with similarity quantified using error rates.

[Agent: Clarify whether the error rates quantify similarity or the performance of the resulting classifications. The current sentence states that similarity itself is quantified using error rates.]

Advances in microscopy have made it possible to move from optical two-dimensional imaging to digital three-dimensional topographic scanning, allowing algorithmic assessment of the similarity between pairs of forensic pattern evidence⁶. This transition has already supported practical advances in bullet and cartridge comparisons, including visual inspection of bullet line scans from three-dimensional topographic images⁷, the use of features extracted by fast Fourier transform (FFT) as the basis of an artificial intelligence system⁸, an automated pilot study based on consecutively matching striae⁹ (a feature also used by some laboratories in casework), and the incorporation of three-dimensional imaging into routine casework at the Alabama Department of Forensic Sciences (ADFS)¹⁰.

When cut wires are found at a crime scene, understanding the characteristics of marks left by bladed tools becomes essential. When a blade cuts a wire, it leaves striations on the surface, as shown in Figure 1. These striations provide a basis for similarity assessment and source identification. Similar scans have supported the development of quantitative methods for analyzing firearms and other toolmark evidence. For example, studies of screwdriver striations have examined key factors affecting comparisons, including angles of attack¹¹, rotational axes¹², and cutting directions¹³. Foundational research on forensic toolmarks has also emphasized data collection and analysis, including the collection and distribution of datasets for bullet and toolmark analysis^{14,15}, numerical methods for improving accuracy and consistency^{16,17}, and techniques for quantifying similarity^{18,19}.

Criminal convictions have relied on cut-wire evidence and related testimony. However, no public database supports assessment of (1) the similarity between striations on wires cut with the same tool or (2) error rates in source identification. Public wire-cut datasets are therefore urgently needed.

Table 1. Structure of available data and files.

	Description	Section
Raw data		
scans/	folder containing 120 topographic 3d scans corresponding to 30 aluminum wire cuts (x3p format)	Cutting wires
meta.csv	meta information for each cut with tool, blade, and location information (CSV format)	Cutting wires
Manual derivatives		
profiles/	folder of files with manually extracted profiles (CSV format)	Extract profiles
Computational derivatives in the folder 'data-derived/'		
wire-signals	signals processed from corresponding profile (zipped CSV format)	Filtered signals
wire_pairwise_ccf	CCF values of all pairwise aligned signals (zipped CSV format)	Align signals
Image files		
pngs/	folder containing pictures of 3d scans of wire cuts (PNG format)	Cutting wires
profile-images/	folder containing pictures of profile extracted from wire cuts (PNG format)	Extract profiles
Visual inventory in the folder 'assessment/analysis-manual/'		
processing-wires	display of pairwise aligned signals from the same sources (HTML format)	Align signals
Technical validation		
variability-assessment/	folder containing related documents for technical validations with the inner folder structure mimicking the structure of the main folder	Technical Validation

In this study, we provide a publicly available dataset of wire-cut scans, develop a systematic pipeline to analyze the data quantitatively, and introduce numerical measures for similarity assessment beyond visual inspection. The dataset includes multiple types of files, as summarized in Table 1.

To ensure full reproducibility, we describe each step in detail, including how we collected and processed the wire-cut samples in Cutting wires, extracted profiles from the scanned wire cuts in Extract profiles, filtered signals from the profiles in Filtered signals, and aligned signals using the cross-correlation function (CCF) as a similarity measure in Align signals. Data Records describes where the dataset is hosted. Technical Validation evaluates the consistency of the assumptions and results and assesses variability. Usage Notes explains how to use the code for signal creation, alignment, analysis, validation, and visualization described in Methods and Technical Validation. Code Availability lists the code provided for reproducibility and future research. We anticipate that this dataset and analysis pipeline will support research on wire-cut evidence, including future studies of real-world crime-scene evidence.

Methods

In this study, we used aluminum wire to create cuts. [Heike: more details: thickness, brand?] The physical properties of aluminum wire make it an excellent material for retaining marks while remaining relatively easy to bend and nontoxic; [Heike: i.e., with a hardness of XXX,] aluminum is soft enough for tools to leave marks but hard enough to resist damage when cut samples are handled under the microscope. In casework, aluminum wire is less common than lead or copper wire. We avoided using lead in the laboratory because of its toxicity. [Heike: XXX look up the hardness value of aluminum, copper, and lead.]

Cutting wires

We used 16-gauge, 1.5-mm anodized aluminum wire to create this wire-cut dataset. For each cut, a 4-inch piece was unspooled and cut with a model KWS-105 Kaiweets wire cutter at one of three blade locations (inner, middle, or outer) to produce one replicate, as shown in Figure 2 (a). Each piece was then cut in half to create 2-inch pieces for each pair of edges, AB and CD. A Sharpie line marked the cut ends, yielding four samples. We then followed the standard scanning protocol for the confocal microscope shown in Figure 2 (b) to scan the wire-tip surfaces. The surfaces were scanned at a resolution of $0.645\mu\text{m} \times 0.645\mu\text{m}$ per square pixel and saved in **x3p** format. Figure 2 (c) shows the AB and CD sides, with C on the reverse of A and D on the reverse of B. We separated the AB and CD sides into two pieces along the bend, resulting in eight scans. We repeated this process for all three blade locations and five wire cutters, with two replicates for each tool-edge-location combination, resulting in a sample of 120 scans. Each piece was labeled according to the following naming convention: T(ool) 1/2/3/4/5, (Edge) A/B/C/D, W(ire), L(ocation) I(nner)/M(iddle)/O(uter), and R(epetition) 1/2. For example, T1AW-LI-R1 denotes the first repetition of a piece cut by tool 1 on edge A at the inner location. To minimize errors in the sampling protocol, two team members cut and labeled the wires together. One team member then scanned the samples in a specified order and named the scans according to the naming convention, and a third team member checked the saved data.

[Agent: The sequence from one 4-inch wire piece to four samples and then eight scans is difficult to reconcile. Clarify what constitutes a cut, piece, sample, edge, and scan, and show explicitly how the procedure yields 120 scans.]

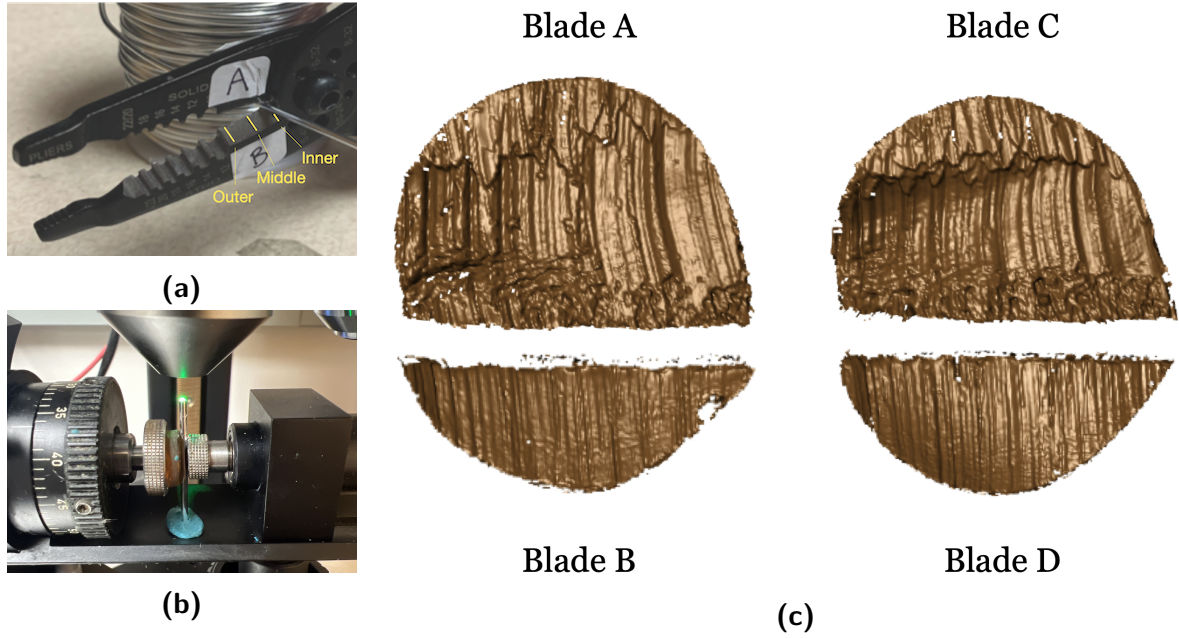


Figure 2. Wire-cut sampling and scanning workflow. (a) A Kaiweets KWS-105 wire cutter with inner, middle, and outer locations marked. (b) A confocal microscope used to scan wire-cut surfaces. (c) Four samples obtained from blades A, B, C, and D after separating the cut edges.

Extract profiles

Numerical comparisons between two replicates cannot be performed directly on the `x3p` files. We must first extract representative profiles from the scans. A profile is a sequence of surface-height values along a user-drawn line. After processing, this profile becomes a signal representing one scan and is used for subsequent comparisons. The profile captures most features of the scan and is drawn orthogonally to the striation marks, which are characterized by variations in groove height. We therefore draw a line across the scan region to capture as much striation information as possible, as shown in dark blue in Figure 3 (a). We then examine the height values along this line. The resulting profile function is plotted in Figure 3 (b) and must be processed before it becomes a useful signal.

Filtered signals

After extracting the profile, we process it to obtain the signal. Two Gaussian filters, as discussed by Cleveland et al.²⁰, are applied to each profile. First, we use a large low-pass filter with a bandwidth of 400 microns to remove the large-scale trend, which would otherwise overwhelm the signal. We then use a small high-pass filter with a bandwidth of 40 microns to average noise and remove spikes. Finally, we remove the extreme tail values and use the resulting signal for subsequent comparisons, as shown in Figure 3 (c).

[Agent: Clarify the filtering operations. A low-pass filter normally retains a large-scale trend rather than removing it, whereas a high-pass filter does not normally average noise. State whether the filtered components are subtracted or whether “low-pass” and “high-pass” refer to different operations here.]

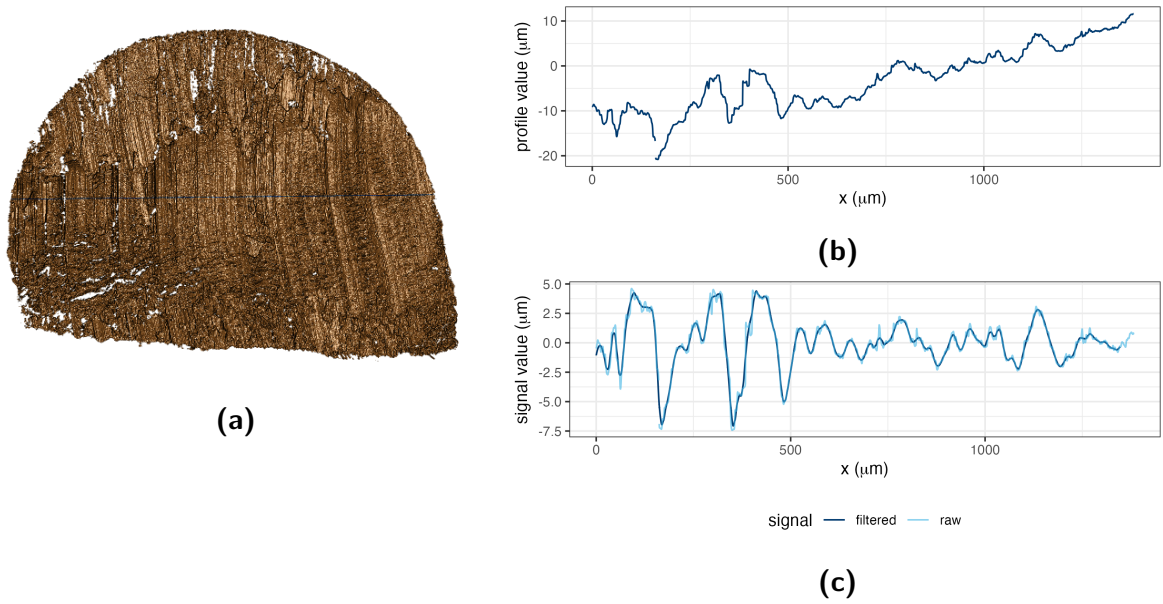


Figure 3. Profile and signal extraction. (a) A dark-blue profile line drawn across the scan striations. (b) The profile function extracted along the line in (a). (c) Raw and filtered signals obtained from the profile function in (b).

Align signals

After extracting signals from different scans, we compare them by maximizing the CCF between each pair to determine their optimal numerical alignment. We compare all possible pairs; for example, the comparison between T1AW-LI-R1 and T1AW-LI-R2 corresponds to a row in Figure 4. We expect signals from two replicates with the same tool-edge-location combination, as shown in the first two columns of Figure 5, to be relatively similar. Their alignments should therefore show substantial overlap and yield high CCF values close to 1. The alignments and values in the rightmost column of Figure 5 meet these expectations.

Data Records

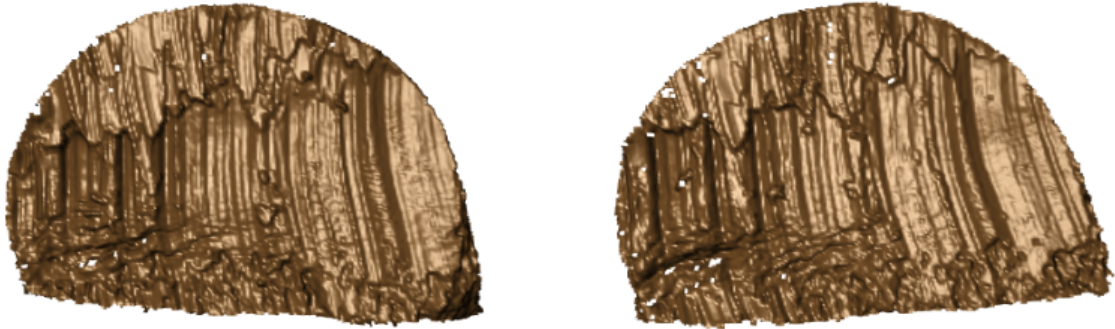
The complete dataset is publicly available to interested researchers through the ISU DataShare repository at <https://iastate.figshare.com/>. The structure of the dataset is described in Table 1. [Agent: Replace the repository homepage with the final dataset landing-page URL or DOI and add a formal data citation before submission.]

Technical Validation

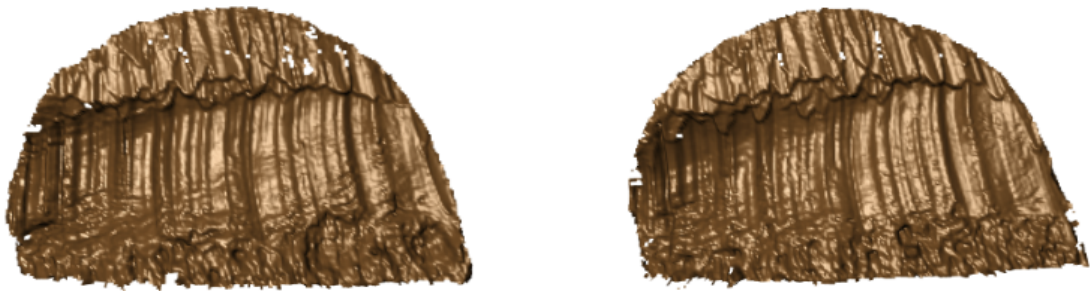
We conducted technical validation in two areas: (1) signal processing and alignment and (2) variability assessment.

We validated scan processing and signal alignment as follows. We expected high CCF values between signals from the same source and low values between signals from different sources. Figure 5, shown earlier in Align signals, meets our expectation for same-source signals. As an example of a different-source comparison, we selected T1AW-LI-R1 and T2AW-LI-R1, shown in Figure 6a and Figure 6b. Their signal alignment, shown in Figure 6c, has a CCF value of 0.2, which is as low as expected. We also summarized the resulting CCFs for all pairwise comparisons in a boxplot and a receiver operating characteristic (ROC) curve, shown in Figure 7 and Figure 8. The ROC curve plots sensitivity, or the true positive rate, against the false positive rate (FPR; $1 - \text{specificity}$). The curve is close to the upper-left corner, indicating

Edge A



Edge C



Edge B



Edge D



Figure 4. Same-source scan pairs. Scans from different sides of tool 1 at the inner location.

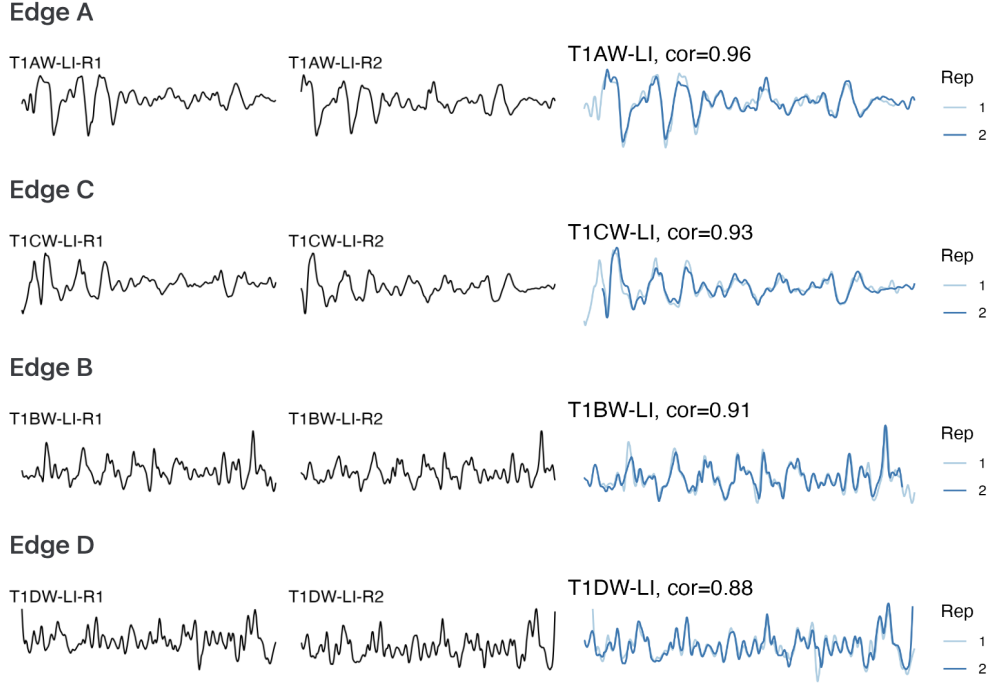


Figure 5. Same-source signal alignment. The first and second columns show the signals extracted from Figure 4, and the third column shows the alignments and CCF values between pairs of signals.

strong classification performance. A threshold of 0.589 controls the FPR below 0.05 with a false negative rate (FNR) of 0, whereas a threshold of 0.658 controls the FPR below 0.01 with an FNR of 0.02. [Tom: (false positive rate (FPR) / false discovery rate (FDR) -> define the H0 or call it false identification rate (FIR)???)] These results are consistent with our expectations and validate our pipeline for processing and aligning signals.

The second validation assessed variability in the pipeline across replicates, stagings, and acquisitions. We selected scans from tool 2 at the middle location for this assessment. Replicate variability was assessed using two different cuts at the same edge and position, as illustrated by the comparison in Figure 5 in Align signals. To assess staging variability, we rescanned replicate 2 under the confocal microscope and introduced the staging label S into the naming protocol. The S1 designation is omitted from labels, so R2 is equivalent to R2-S1. For all four edges of tool 2 at the middle location, we created three stagings (R2, R2-S2, and R2-S3), computed pairwise CCFs between stagings, and calculated their mean and standard deviation (SD).

Within each staging, we further examined the effect of different acquisitions by keeping the samples on the confocal microscope while varying the lighting conditions. We introduced A as the acquisition label. For edge A at staging 3, we performed four acquisitions and omitted the A1 designation from the first acquisition label, so R2-S3 is equivalent to R2-S3-A1. For the other three edges, we conducted two acquisitions at staging 3, denoted by R2-S3 and R2-S3-A2. We again computed pairwise CCFs between stagings and calculated their mean and standard deviation.

[Agent: This paragraph describes acquisition variability, but its final sentence says that CCFs were computed between stagings. Confirm whether “stagings” should be “acquisitions.”]

With this setup, we expected CCF values to become more consistent from replicate to staging to acquisition as the scanning conditions became progressively more stable. Consequently, variability should decrease along this sequence. We also compared replicates, stagings, and acquisitions, expecting smaller differences between acquisitions and stagings than between stagings and replicates. We visualized the results as scatter plots in Figure 9, which also shows

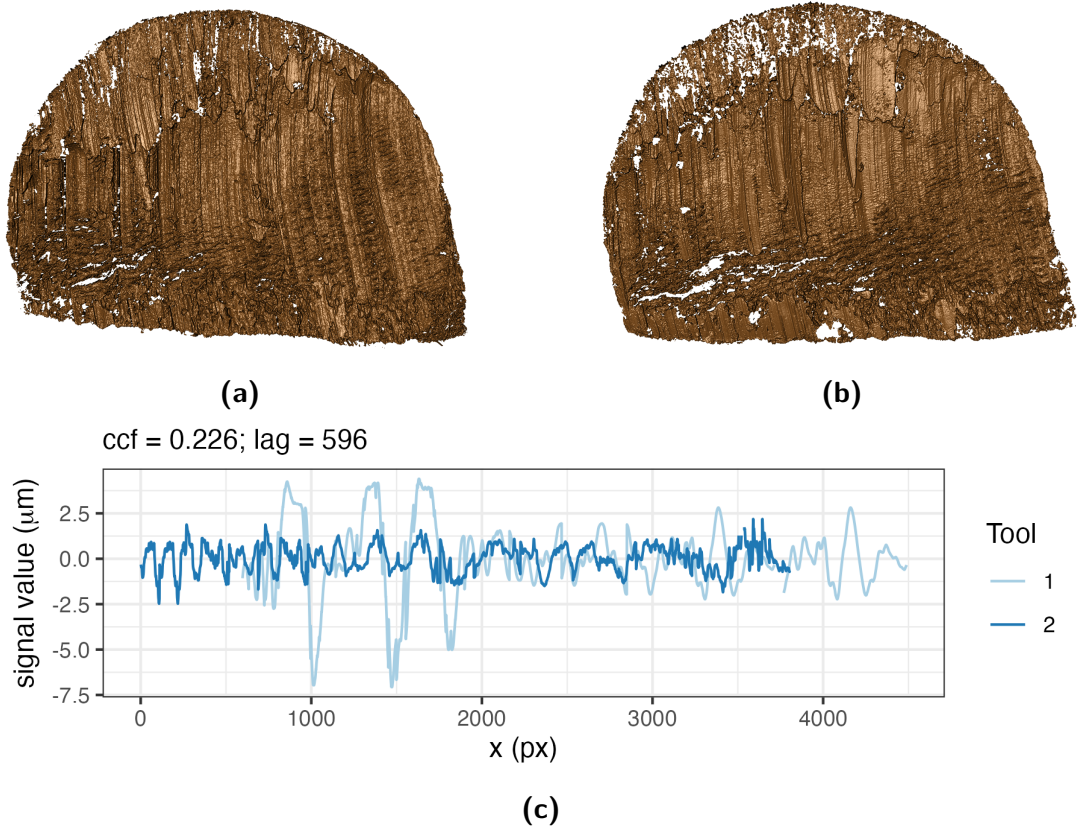


Figure 6. Different-source scan and signal comparison. (a) Scan T1AW-LI-R1 cut by tool 1. (b) Scan T2AW-LI-R1 cut by tool 2. (c) Alignment of signals from T1AW-LI-R1 and T2AW-LI-R1.

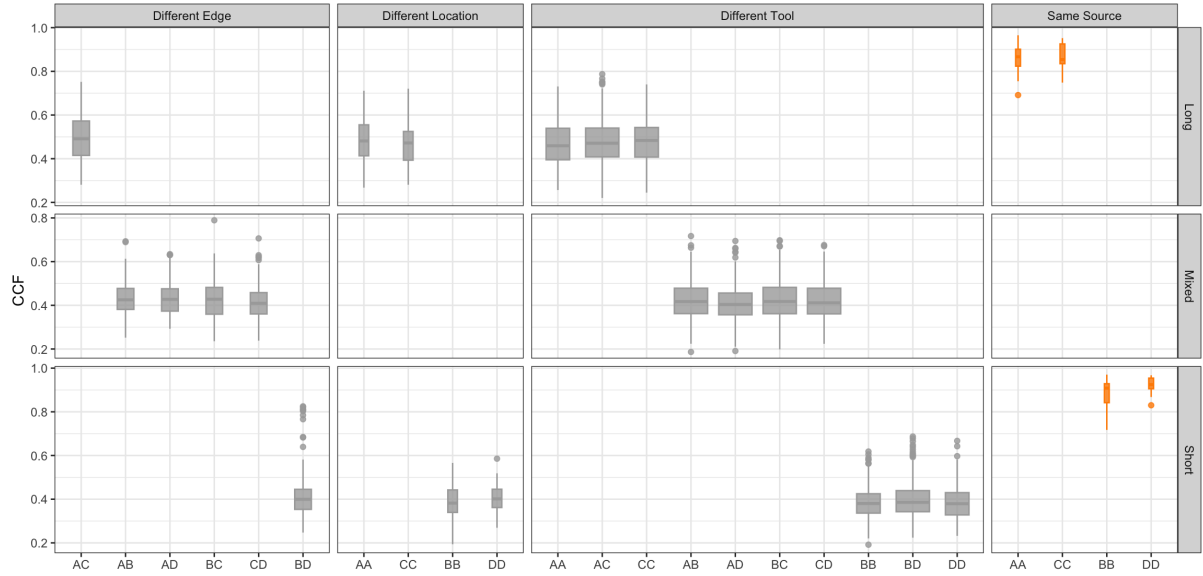


Figure 7. Same- and different-source CCF distributions. Signals from the same sources have higher CCFs than those from different sources.

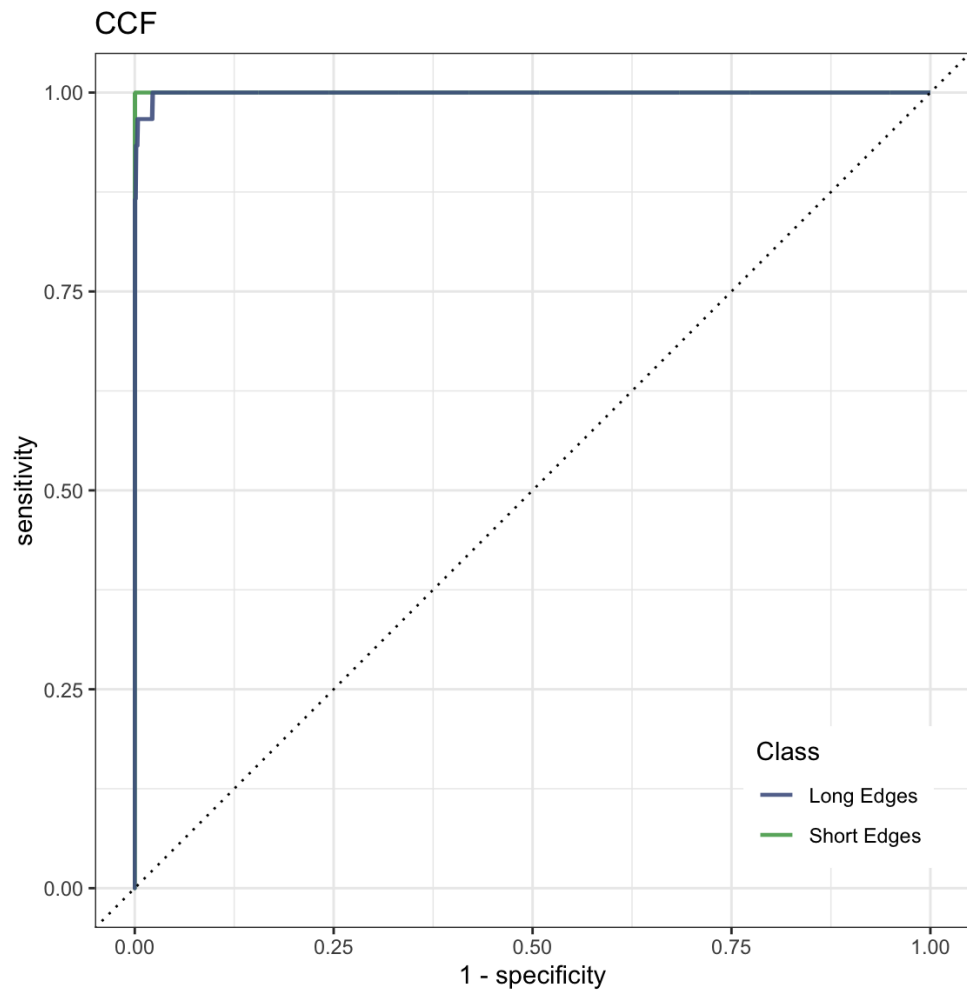


Figure 8. Classification performance based on CCF values. The ROC curve bends close to the upper-left corner.

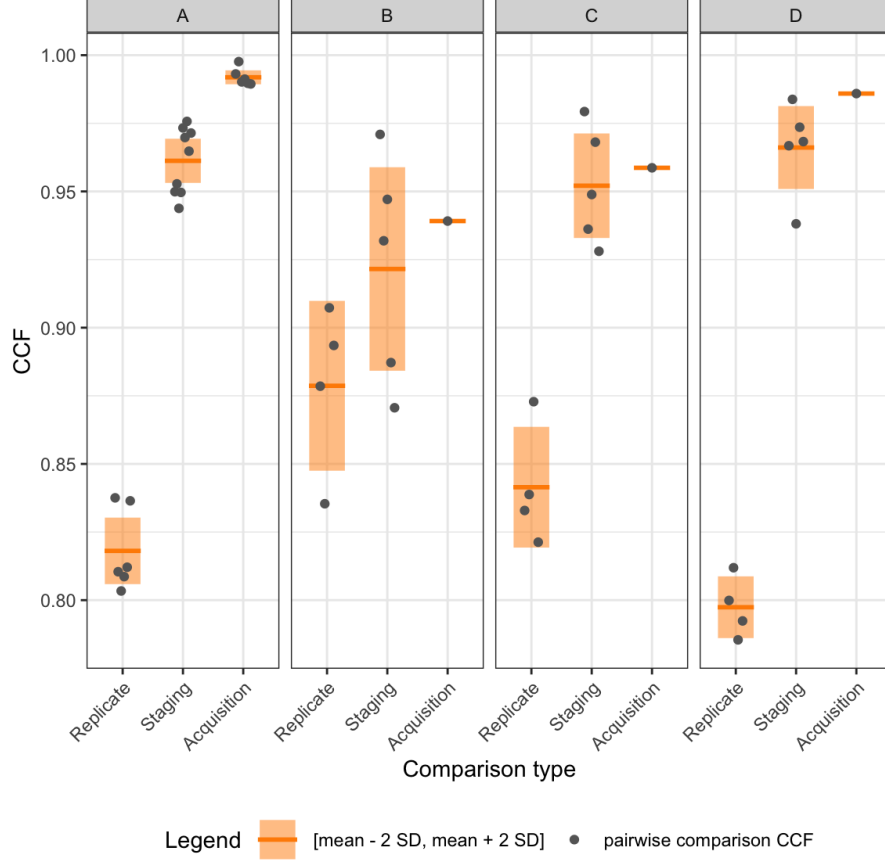


Figure 9. Variability across scanning conditions. Scatter plots of CCF values across replicate, staging, and acquisition conditions, with means and two-standard-deviation intervals.

intervals spanning two standard deviations. The higher orange bars indicate an increase in the mean, and the narrower intervals indicate a decrease in the standard deviation. The detailed numerical results, including means, standard deviations, and differences between means, are shown in Table 2 and confirm our expectations.

Our variability assessment does not include variation in CCF values caused by differences in the manual extraction of profiles from scans. To support reproduction of the results reported here, we have included the extraction locations and values for each scan.

Table 2. Comparison of CCF means and standard deviations across different replicates, stagings, and acquisition settings.

Edge	Replicate (R)		Staging (S)			Acquisition (A)		
	Avg_R	SD_R	Avg_S	SD_S	$\text{Avg}_S - \text{Avg}_R$	Avg_A	SD_A	$\text{Avg}_A - \text{Avg}_S$
A	0.818	0.015	0.961	0.012	0.143	0.992	0.003	0.031
B	0.879	0.031	0.922	0.042	0.043	0.939	—	0.017
C	0.841	0.022	0.952	0.021	0.111	0.959	—	0.007
D	0.797	0.011	0.966	0.017	0.169	0.986	—	0.020

Table 3. Overview of available code.

	Description	Section
Inspect raw scans		
1-create_pngs_from_x3p.R	obtain images of x3ps in scans/	Cutting wires
Extract profiles		
2-create_profiles_from_x3p.R	manually extract profiles from each scan	Extract profiles
3-create-single-profile-file.R	create meta profile information	Extract profiles
Derive signals		
4-create_signals_from_profiles.R	derive signals from each profile	Filtered signals
Align signals		
5-create-images.R	create images for pairwise alignment	Align signals
6-align-pairwise.R	compute pairwise alignment CCF values	Align signals
7-all-comparison-results.R	visualize comparison results	Align signals

Usage Notes

The R package `x3ptools`²¹, available from CRAN, supports working with files in `x3p` format. Sample R scripts for converting `x3p` scans into signals and aligning those signals are available in the `assessment/code` folder of the GitHub repository `heike/wirecuts-data`, as described in Table 3.

We conducted pairwise comparisons and visualized some of the results in Align signals and Technical Validation. Here, we present additional analysis plots. When the CCF values are displayed in a tile map across tools, locations, and edge combinations, we expect only the diagonal to contain high values close to 1, shown in orange, because the diagonal represents same-source comparisons. We expect the rest of the matrix to contain low values close to 0, shown in gray. The pattern in Figure 10 is generally consistent with this expectation, except for a few cases involving edge D of tool 5. The density plot in Figure 11 shows the distributions of CCF values for same-source and different-source comparisons. The overlap between the tails of these distributions provides a rough threshold for drawing conclusions.

[Agent: Scientific Data limits Usage Notes to practical reuse guidance and does not allow analysis or conclusions there. Decide whether to remove this analysis paragraph and Figures 10–11 or move them to the optional Data Overview section, which is limited to one paragraph and one or two figures or tables.]

Data Availability

[Agent: A short Data Availability statement is mandatory. Repeat the final repository URL or DOI and the principal accession or dataset identifiers already described in Data Records.]



Figure 10. Pairwise CCF tile map. Signals from the same source have CCFs close to 1.

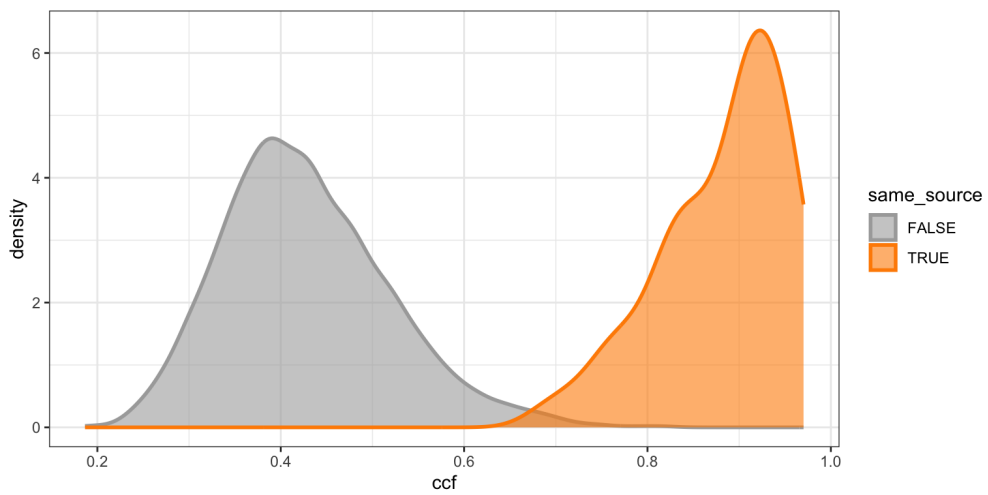


Figure 11. CCF density distributions. The overlapping distribution tails can be used as a rough threshold for drawing conclusions.

Code Availability

We made available all code used to inspect raw scans, extract profiles, derive signals, align signals, and visualize the comparison results discussed in Methods and Technical Validation. The files are described in Table 3. All results can be reproduced using the provided code.

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Author Contributions

[Agent: Confirm whether to use the CRediT taxonomy for the Author Contributions statement; Scientific Data does not require the linked Elsevier wording.]

Y.L.: Methodology, Software, Validation, Data Curation, Writing: Draft; H.H.: Conceptualization, Methodology, Validation, Writing: Review and Editing; C.M.: Laboratory Supervision; E.A.: Physical Specimen, Scanning; J.S.: Forensic Advice; A.C.: Funding Acquisition.

All authors reviewed the manuscript.

Competing Interests

H.H. is a technical advisor to the Association of Firearm and Toolmark Examiners (AFTE), a fellow of the American Statistical Association (ASA), and a member of the ASA Forensic Science Committee. H.H. has testified as a court witness on behalf of Judge April Neubauer in the New York State Supreme Court, Criminal Term, in New York City. [Heike: other competing interests - Alicia?]

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